TrustSphere AI — Digital Evidence Verification Report

Generated 2026-10-23 16:43:06 UTC · Case ID 99a217a9-0b95-46c7-80d8-738896f3a25a

1. Evidence Identification

Original filename	9_photo_matched_AUTHENTIC.jpg
File type	image
SHA-256 hash	`044903146fb42e043d6581d304cecde7d0860f137e26175583a8a899bc988ed7`
Uploaded (UTC)	2026-10-23T16:43:06.654142
Analysis status	done

2. Forensic Verdict

INCONCLUSIVE — overall authenticity score 0.52 (0 = fake, 1 = authentic)

Manipulation type	Spliced / Copy-Move Forgery (confidence 0.59)

Reasoning:
• Overall authenticity score 0.52 → Inconclusive.
• Deep classifier flagged synthetic imagery (p=0.58).
• Error-level analysis shows inconsistent compression regions (splice).

3. Agent Findings

metadata — authenticity 0.450

Check	Status	Note
camera_info	suspicious	No camera make/model — common in AI-generated or re-encoded images
software	ok	No editing software tag
timestamps	suspicious	No EXIF timestamps present

image_forensics — authenticity 0.562

Check	Status	Note
ela	suspicious	Mean error level 17.71 (heatmap saved)
fft	ok	High-frequency energy ratio 0.647
deep_model	suspicious	Deepfake classifier authenticity 0.419
xai_gradcam	ok	Model-attention heatmap saved (gradcam)

security — authenticity n/a (no signal)

Check	Status	Note
c2pa_manifest	ok	No C2PA content credentials embedded (typical for most media).
invisible_watermark	ok	No known AI-generator watermark detected.

4. Explainability Exhibits (XAI)

Exhibit — metadata feature importance (exact Shapley contributions; green → authentic, red → fake)

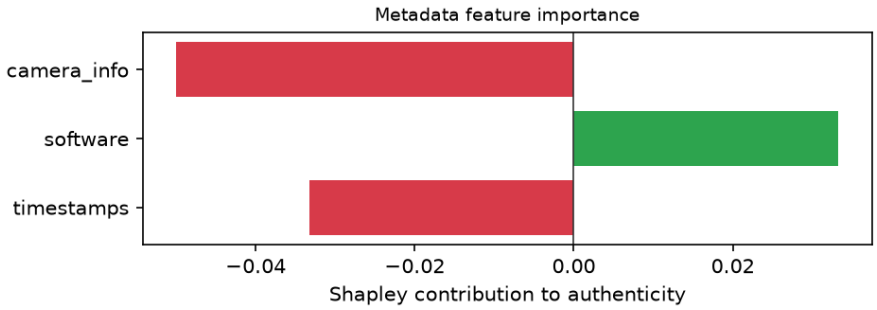
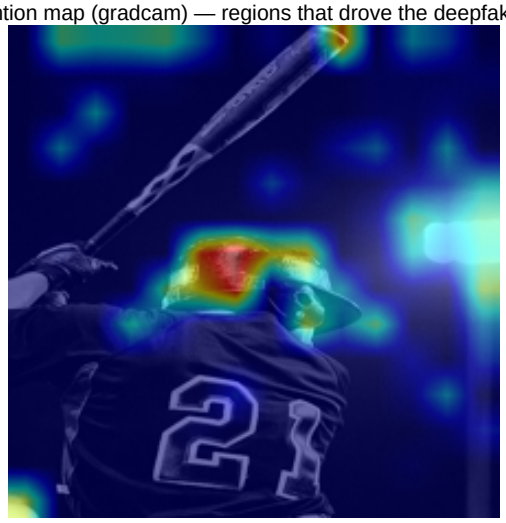


Exhibit — image_forensics: ELA heatmap (bright regions compress inconsistently → possible splice)



Exhibit — image_forensics: model attention map (gradcam) — regions that drove the deepfake classifier's decision



5. Chain of Custody

Event	Timestamp (UTC)	Detail
uploaded	2026-10-23T16:43:06.654142	
analyzed	2026-10-23T16:43:06.969989+00:00	
signed	2026-10-23T16:43:06.970124+00:00	`6217864d3e24af8f2157b4800b3b29c95f497c141297c36aae97212c7b77f184`
blockchain_skipped	2026-10-23T16:43:06.970138+00:00	`POLYGON_PRIVATE_KEY not set — anchoring disabled`

Blockchain network	polygon-amoy

Anchor status	skipped

6. Record Integrity — Digital Signature

The analysis record below is signed with the TrustSphere server's Ed25519 key. Any alteration of the signed payload invalidates the signature (verifiable via GET /api/verify/{file_id}).

Algorithm	Ed25519
Key fingerprint	`6217864d3e24af8f2157b4800b3b29c95f497c141297c36aae97212c7b77f184`
Signature	`b7ece9e8704ff54b97325c58198694c5feecb4f687694cc04dc97c896f4438c9455962aff9 2c7ac71b1f68b7ea5487c3a65a3b59daefc6f42f4717c633f3a806`

This report was generated automatically by TrustSphere AI. The verdict is a statistical forensic assessment, not a legal determination.